

PRODUCT REQUIREMENTS DOCUMENT

Boojho & Paheli’s Fraction Mela

Grade 5 Fractions Applet Suite

Complete PRD covering concept flow, interactions, misconception handling, feedback rules, student Q&A, analytics, accessibility, and rollout guidance

Prepared for: Interactive applet design and development

Source basis: Grade 5 Fractions chapter activity flow

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1. Executive Summary

This document defines a complete product requirements document for a Grade 5 fractions applet suite built around Boojho and Paheli. The suite is designed to convert the full chapter flow into a sequence of interactive, misconception-aware learning experiences.

The product goal is not only to make learners practice tasks, but to help them understand why a concept works, how to test their own thinking, and how to recover from common errors with visual support.

Product	Boojho & Paheli’s Fraction Mela
Audience	Primary: Grade 5 learners Secondary: teachers, parents, facilitators
Core promise	Every concept applet should let the child try, ask, see, correct, and retry.
Chapter scope	Fractions as parts of a whole; same whole comparison; equivalent fractions; comparison with same denominator and same numerator; fractions greater than 1; number-line representation; comparison using 1 and 1/2 as benchmarks.

2. Product Goals and Design Principles

- Turn chapter activities into highly interactive scenes without losing the original conceptual progression.
- Catch likely misconceptions in the moment and respond with constructive visual correction.
- Allow the child to ask questions at any point and receive a short explanation, a visual explanation, and a rule-based explanation.
- Use Boojho as the natural question-asking character and Paheli as the calm explanatory guide.
- Make every applet usable in classroom, guided-home, and self-paced contexts.

3. Cross-Applet Interaction Model

Each applet follows a shared interaction grammar so the experience feels predictable for the learner and easier to implement consistently.

Persistent control	What it does	Expected response
Ask Boojho	Lets the learner tap, choose, or type a question.	Opens the question-answer layer with child-friendly prompts.
Paheli Explains	Provides a concise explanation using the current screen.	Shows a short answer followed by visual reasoning.
Show Me	Replays the concept through animation.	Demonstrates the idea step-by-step.
Why Is This Wrong?	Explains the learner's specific mistake.	Names the error, shows why it fails, and gives a retry.
Try an Easier One	Offers a simpler version of the same idea.	Builds confidence before returning to the main task.
Tell Me the Rule	States the general rule after the learner sees the model.	Connects visual understanding to formal language.

4. Common Feedback and Misconception Framework

Every concept applet must support the same instructional response cycle:

- Detect the learner action.

- Identify the likely misconception or gap.
- Respond in positive language.
- Show a visual correction on the same representation.
- Offer one simpler example if needed.
- Return the learner to a retry state.

Recommended response order: short answer -> visual explanation -> simple rule
-> check-back question.

5. Applet 1 — Boojho's Chocolate and Grid Mystery

Purpose	Teach that fractions are parts of a whole and that comparison is valid only when the wholes are the same. The applet also covers grid-based fraction identification and shading tasks.
Chapter coverage	Chocolate comparison scene; identifying $\frac{1}{2}$ and $\frac{1}{3}$; shading $\frac{1}{8}$, $\frac{1}{6}$, and $\frac{1}{12}$ on different grids; locating $\frac{1}{3}$ on the grids.
Learning goals	• Recognise fractions as equal parts of a whole. • Understand that fraction comparison requires the same whole. • Represent simple fractions accurately on area models and grids.

Key screens

Screen	Intent
Story intro	Boojho makes a surprising comparison claim.
Chocolate comparison board	Learner explores different-sized wholes.
Same-whole correction mode	Learner aligns wholes and compares fairly.
Grid shading zone	Learner shades target fractions and marks $\frac{1}{3}$.
Reflection screen	Learner explains the rule in simple words.

User flow

1. Observe Boojo's claim and choose whether it seems correct.

2. Manipulate the size of the chocolates and see how the meaning of $\frac{1}{2}$ and $\frac{1}{3}$ changes with the whole.
3. Use the same-whole view to make both chocolates equal in size and compare again.
4. Shade the requested fractions on three different grids.
5. Mark where $\frac{1}{3}$ can be seen and complete the rule check.

Misconceptions, hints, and feedback

Misconception to catch	Hint layer	Feedback / correction
$\frac{1}{2}$ is always greater than $\frac{1}{3}$.	Check whether both wholes are the same size first.	The app highlights that the wholes are different and blocks final comparison until they match.
Fractions can be compared even when wholes differ.	Tap the whole outline to see what the fraction belongs to.	Paheli overlays equal wholes and shows how the comparison changes once the same unit is used.
Unequal parts can still represent a fraction fairly.	Turn on equal-part guides before shading.	If parts are unequal, the app marks them as invalid and explains why fractions need equal parts.

Sample dialogue

Boojho: I thought $\frac{1}{2}$ must always be bigger than $\frac{1}{3}$.

Paheli: Only if both fractions come from the same whole. Let us check the chocolates first.

Child: Why is $\frac{1}{3}$ bigger here?

Paheli: Because this $\frac{1}{3}$ is from a larger whole. Now try making both wholes equal.

Success criteria

- Learner states that fractions must be compared using the same whole.
- Learner correctly shades all target fractions.
- Learner correctly rejects unequal partitions as invalid fraction models.

6. Applet 2 — Paheli's Fraction Kit Workshop

Purpose	Teach how different fractional pieces can make a whole and how equivalent fractions can be discovered through physical composition and decomposition.
Chapter coverage	Making a whole with $\frac{1}{5}$ pieces; using $\frac{1}{2}$ and two $\frac{1}{4}$ pieces; checking $\frac{1}{3} = \frac{2}{6}$; finding how many small pieces make larger fractions; creating wholes with specified combinations.
Learning goals	• Build wholes from fraction pieces. • Understand denominator as the number of equal parts in the whole. • Recognise equivalent fractions through exact coverage of the same whole.

Key screens

Screen	Intent
Fraction tray	All pieces are available for drag-and-drop exploration.
Build-a-whole board	Learner assembles one complete whole.
Overlay matcher	Learner tests equivalent pieces such as $\frac{2}{4}$ and $\frac{1}{2}$.
Challenge board	Learner solves kit-based prompts.
Discovery notebook	Equivalent relations and successful wholes are recorded.

User flow

1. Drag pieces from the tray and fill one whole.
2. Try building a whole using different constraints such as only $\frac{1}{5}$ pieces or a mix of $\frac{1}{2}$ and $\frac{1}{4}$ pieces.
3. Overlay pieces to test whether two expressions cover the same amount.
4. Solve mini-prompts such as how many $\frac{1}{6}$ pieces make $\frac{1}{3}$.
5. Review discoveries in the notebook and verbalise the pattern.

Misconceptions, hints, and feedback

Misconception to catch	Hint layer	Feedback / correction
More pieces means more quantity.	Try placing one fraction directly on top of the other.	The app shows exact overlay matches for equivalent fractions.
A larger denominator means a larger fraction.	Tap the denominator to see how many equal cuts the whole has.	If the learner thinks a larger denominator means a larger part, the unit piece visibly shrinks as the denominator increases.
Equivalent fractions are different because they use different numbers.	Use the split tool to break a larger piece into equal smaller pieces.	If the whole is overfilled or underfilled, gaps or overlap are shown immediately.

Sample dialogue

Child: Why is $\frac{2}{4}$ equal to $\frac{1}{2}$?

Paheli: Because they cover the same amount of the same whole. Watch them line up.

Boojho: But there are more pieces in $\frac{2}{4}$.

Paheli: Yes, but each piece is smaller. More pieces does not always mean more quantity.

Success criteria

- Learner builds one whole in at least three different ways.
- Learner identifies at least three equivalent fraction pairs correctly.
- Learner explains that a larger denominator creates smaller unit parts.

7. Applet 3 — Equivalent Fraction Pattern Studio

Purpose	Teach learners to generate and verify equivalent fractions using both visual partitioning and numerical pattern recognition.
Chapter coverage	Partitioning $\frac{1}{3}$ into $\frac{2}{6}$, $\frac{3}{9}$, and $\frac{4}{12}$; extending equivalent-fraction patterns for $\frac{1}{3}$ and $\frac{2}{5}$; filling blanks; ticking equivalent pairs; completing missing numerator and denominator values.
Learning goals	• Generate equivalent fractions by changing numerator and denominator in the same way. • Link visual partitioning with number patterns. • Distinguish equivalent pairs from non-equivalent distractors.

Key screens

Screen	Intent
Pattern wall	Shows starting models such as $\frac{1}{3}$ and $\frac{2}{5}$.
Partition builder	Learner draws additional equal cuts.
Fraction machine	Learner chooses a multiplication factor.
Repair station	Learner fills in missing values.
Rule recap	Learner states how equivalent fractions are formed.

User flow

1. Start from a fraction model already shown on the wall.
2. Add more equal partitions and observe the new fraction name.
3. Use the fraction machine to multiply numerator and denominator by the same factor.
4. Solve missing-value and equivalent-pair tasks.
5. Summarise the pattern in words before exiting.

Misconceptions, hints, and feedback

Misconception to catch	Hint layer	Feedback / correction
Only the numerator changes in an equivalent fraction.	Change top and bottom together.	If only one value changes, the app shows the shaded area no longer matches.
Only the denominator changes in an equivalent fraction.	Use the same factor on both numbers.	If the learner uses addition without preserving the value, the app demonstrates the mismatch through overlay.
Adding the same number always gives an equivalent fraction.	Check whether the shaded amount stayed exactly the same.	Correct answers trigger both a visual match and a stated number pattern.

Sample dialogue

Child: Can I just add 1 to both numbers?

Paheli: Let us test it. If the shaded amount changes, the fractions are not equivalent.

Boojho: $\frac{3}{9}$ looks different from $\frac{1}{3}$.

Paheli: The names differ, but the amount is the same. That is what makes them equivalent.

Success criteria

- Learner can generate equivalent fractions using a common factor.
- Learner can reject non-equivalent pairs with a reason.
- Learner states that equivalent fractions name the same amount of the same whole.

8. Applet 4 — Boojho and Paheli's Compare-the-Bite Café

Purpose

Teach comparison rules for fractions with the same denominator and with the same numerator using familiar food contexts and symbolic comparison signs.

Chapter coverage

Comparing fractions with the same denominator; comparing fractions with the same numerator; food-based comparison scenes; use of $<$ and $>$ signs.

Learning goals

- Compare fractions with the same denominator by counting equal-sized parts.
- Compare fractions with the same numerator by comparing unit-part size.
- Select the correct rule before applying it.

Key screens

Screen	Intent
Café intro	Learner enters a food-sharing scene.
Same denominator counter	Bars and food plates show equal-sized parts.
Same numerator counter	Piece size changes while numerator stays fixed.
Rule check	Learner chooses which rule applies.
Speed round	Learner solves quick comparisons with reasons.

User flow

1. Observe two fractions and first identify what remains the same.
2. Compare same-denominator pairs by counting pieces.
3. Compare same-numerator pairs by evaluating piece size.
4. Drag the correct comparison sign into place.
5. Give or select a reason for the answer.

Misconceptions, hints, and feedback

Misconception to catch	Hint layer	Feedback / correction
A larger denominator always means a larger fraction.	Ask first: what is the same here?	The app highlights equal-sized parts for same-denominator pairs and counts them visually.
A larger numerator always means a larger fraction in every situation.	If denominators are the same, count how many equal parts are taken.	For same-numerator pairs, the app shows that fifths are larger than sixths and similar contrasts.
The learner can compare by looking only at the biggest visible number.	If numerators are the same, compare the size of each part.	If the learner applies the wrong rule, the app pauses and redirects them to identify what is fixed first.

Sample dialogue

Child: Why is $\frac{4}{6}$ less than $\frac{4}{5}$?

Paheli: Because both have 4 parts taken, but sixths are smaller than fifths.

Boojho: I just saw a 6 and thought it was more.

Paheli: That is a common mistake. Here the part size matters more than the larger number.

Success criteria

- Learner consistently chooses the right comparison rule before answering.
- Learner correctly compares same-denominator and same-numerator pairs.
- Learner explains at least one comparison in words, not only with symbols.

9. Applet 5 — Fraction Journey Beyond One and Benchmark Detectives

Purpose	Teach fractions greater than 1, connect them to mixed numbers and number lines, and help learners compare fractions using 1 and $\frac{1}{2}$ as benchmarks.
Chapter coverage	Using halves, fourths, and thirds to build fractions beyond 1; number-line representations; identifying fractions greater than 1; comparison using 1 as a reference; identifying fractions equal to and less than $\frac{1}{2}$; comparison using $\frac{1}{2}$; repeated $\frac{1}{4}$ ant-length challenge; final bonus riddle.
Learning goals	• Represent fractions greater than 1 using concrete models and number lines. • Bundle improper fractions into wholes and leftover parts. • Use 1 and $\frac{1}{2}$ as quick comparison benchmarks.

Key screens

Screen	Intent
Family feast intro	Food-sharing story launches the applet.
Beyond one board	Learner builds improper fractions with pieces.
Number-line hop zone	Learner places fractions using equal hops.
Benchmark sorter	Learner sorts fractions around $\frac{1}{2}$ and 1.
Ant march challenge	Learner repeats $\frac{1}{4}$ units on a line.
Bonus riddle	A light end-screen reasoning task appears.

User flow

1. Build values such as $\frac{5}{2}$ and $\frac{9}{4}$ using fraction pieces.
2. Watch the app bundle complete wholes automatically.
3. Place the same values on a number line using equal hops.
4. Sort fractions into less than, equal to, or greater than $\frac{1}{2}$ and 1.
5. Complete the repeated-unit challenge and exit check.

Misconceptions, hints, and feedback

Misconception to catch	Hint layer	Feedback / correction
Fractions must always be less than 1.	Bundle full wholes first.	If the learner rejects an improper fraction, the app converts the pieces into wholes and leftover parts visibly.
A fraction with numerator larger than denominator is impossible or wrong.	Count equal hops on the number line.	If the learner misplaces a point on the number line, the app shows equal hop intervals from zero.
More than $\frac{1}{2}$ also means more than 1.	Compare against $\frac{1}{2}$ and 1 before trying full calculation.	If the learner confuses $>\frac{1}{2}$ with >1 , the benchmark sorter shows both reference points side by side.

Sample dialogue

Child: How can $\frac{5}{2}$ be real?

Paheli: It means 5 halves. Two halves make one whole, so 5 halves make two wholes and one half.

Boojho: If it is more than half, is it more than one?

Paheli: Not always. Let us compare it with $\frac{1}{2}$ first and then with 1.

Success criteria

- Learner correctly represents improper fractions as wholes plus leftover parts.
- Learner places fractions accurately on a number line.
- Learner uses $\frac{1}{2}$ and 1 as helpful reference points in comparison tasks.